

Lesson 2.1: Building a Model

Location: Classroom

Grade Level: High School

Time: Two class sessions of 50 minutes

Driving Question

- How will climate change affect what we value about Southern California's ocean?

Lesson Synopsis

During this 60-minute session, students work together in small teams to develop a model showing their initial ideas about how climate change will affect Southern California's ocean.

Drawing from their analysis of community interviews, student teams brainstorm the components, processes, and human interactions that play a role in the ocean. They review their interview analysis and select one idea that their community values about the ocean to focus on. They then use SageModeler to begin developing a model that shows how they predict that climate change might affect the value. Finally, they reflect on how they can use generative AI chatbots to identify additional perspectives and ideas to incorporate into their model.

What students will figure out:

- Climate change affects ocean systems, which in turn impacts the communities that rely on them.
- Different individuals within a community place different values on the ocean system, such as emotional value, financial value, or even traditional value.
- Scientific models help us predict and understand these interactions and how changes to a system may impact the values of individuals in our communities

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Generate a list of components and processes involved in climate change's impact on Orange County's ocean.	Student lists created during the <i>Putting the pieces together</i> section.
2. Use SageModeler to create a static model that shows their initial ideas about how climate change will impact something they value connected to the ocean.	Student models created during the <i>Putting the pieces together</i> section.
3. Reflect on how they can identify areas of improvement for their model using generative AI.	Class discussion during the <i>Problematize</i> section.

Building towards NGSS and AI Literacy:

- **HS-ESS3-6:** Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. In this lesson, students will construct a model based on local climate change and how human activity contributes to positive and negative effects in the environment.

Lesson Overview

Section	Time	What Happens
Navigate	15 minutes	Students reflect on the similarities between values that they share and values that communities share connected to the ocean.
Investigate	50 minutes	First, students work in their teams to brainstorm the components, processes, and interactions connected to how climate change will affect the ocean system. Next, student teams then select a value connected to the ocean. They choose 10-12 items from their list that are most connected to their value to prioritize including in their model. Finally, the teams then use SageModeler to develop a model showing how climate change will affect their chosen value connected to the ocean.
Navigate	25 minutes	Students conduct a virtual gallery walk to compare their models with the models developed by other groups. They discuss any patterns that they notice and identify whether there are any additional ideas they might add to their model.
Problematize	10 minutes	Students begin reflecting on their models and what additional perspectives they might incorporate to improve the models.

MATERIALS

- [Lesson 2.1 Slideshow](#)
- [Science Journal](#)
- [SageModeler template](#)

BEFORE YOU START TEACHING

1. Review [Student SageModeler Instructions](#) to model or support students.
2. Include a numbered list of general steps to set up the investigation before students arrive.
3. This can also include other things that educators should do or think about before starting the lesson, such as checking for safety concerns.

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: In this lesson, students will use the information they have gathered from the previous module to create a model of climate change's impact on their communities. It will include factors that affect humans and the environment and how they create a cycle of both negative and positive effects.

- **HS-ESS3-6:** Construct a model based on local climate change and how human activity contributes to positive and negative effects in the environment.

Students will encounter the terms: **components**, **processes**, and **interactions** as part of building their systems models. They will use the previously gathered information to identify patterns in human activity, values, and their effects on the environment. It might also be natural for students to incorporate other concepts they might have encountered previously, such as **carbon cycle**, **carbon sinks**, **sea level rise**, **changing weather patterns**, and **climate mitigation and adaptation**.

Where We Are NOT Going: Similar to Module 1 lessons, this lesson focuses on local community perspectives and impact of climate change on Southern California's ocean. Students will not investigate the global impacts of climate change in depth. While they are encouraged to incorporate other concepts linked above (e.g., carbon cycle, carbon sinks, sea level rise, rising sea temperature, climate mitigation and adaptation) into their model, we will examine these concepts in more depth in Module 3 (focusing on the chemistry of climate change).

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: Reflecting on What We've Done So Far (15 minutes)

1. Open the slideshow to **Slide 1** and inform students that today is the beginning of module 2. Tell students that in module 2, they will work on transferring the information gathered in module 1 to create a model on how communities value the ocean.
2. Advance the slideshow to **Slide 2** and invite the class to reflect on what they have accomplished in the last module
 - What did you do in module 1?
 - What kind of information did you gather?
3. Advance to **Slide 3** to support student responses and review that in the previous module. In Module 1, you:
 - Considered how the ocean is changing
 - Conducted research on how climate change will impact Southern California's ocean
 - Interviewed community members to see how changes to the ocean will impact what we value
4. Move forward to **Slide 4** to share with the class the lesson 2.1 overview.
5. When you're ready, advance to **Slide 5**. Ask students to think back to their personal values connected to the ocean, as well as the reasons that community members care about the ocean that were shared during the interviews.

Have student groups reflect on the following questions:

- How do you interact with the ocean? Why do you value it?
 - How does your community interact with the ocean? Why do they value it?
 - What impact do you think climate change will have on the way communities interact with the ocean?
6. When student groups are done discussing, invite the class back together and have students share out.

7. Inform students that today, they will take the information that they gathered and develop a model using their initial ideas of how climate change will affect Southern California's ocean.

INVESTIGATE: Brainstorming What to Include in Our Model (20 minutes)

1. Ask student teams from module 1 to get together for a brainstorming session. Advance to **Slide 6**, and invite students to think about the following question: **What are the components and processes that make up the ocean?**
2. Ask student teams to open up their **Science Journal** to **Module 2.1: Building a Model (Student Task: Brainstorming Components, Processes, & Interactions)** (p.9) where they will see a brainstorming document.
3. Provide students time to think about **only the components and processes** that they think make up the ocean and add their ideas to the brainstorming document.
 - **Components:** Individual elements that make up the ocean system, which can include both living (biotic) and non-living (abiotic) elements, such as organisms like fish or kelp, water, oxygen, and nutrients.
 - **Processes:** Actions in the ocean system that occur over time, such as tidal shifts, storm events, or photosynthesis.
4. After students have had time to brainstorm their components and processes, advance to **Slide 7** and guide students to start thinking about the different interactions humans may have with the ocean system:
 - **Interactions:** The ways in which humans and different components of the ocean system engage with one another. These interactions could be positive, negative, or neutral, such as beach clean-ups, emitting carbon, or surfing.
5. Give student teams time to brainstorm their interactions and add them to their brainstorming document in their science journal.
6. Once students have their brainstormed list of components, processes, and interactions, advance to **Slide 8**. Tell students that their next step in the modeling process is to select a value that is connected to the ocean. Invite them to think back to when they analyzed the community interviews and respond in their **Science Journal to Module 2.1: Building a Model (Student Task: Brainstorming Components, Processes, & Interactions)**, p. 9-10.

- Based on your interview responses, what evidence shows why people care about the ocean? Include specific quotes or examples from your interview to support your answer (reference [Module 1.2 Part 2: Conducting Community Interviews notes](#)).
 - How did your team categorize these values or themes?
 - Which value from your interviews will your team include in your model, and how does this value connect to the ocean system or the impacts of climate change?
7. Advance to **Slide 9** and give students a few minutes to choose a value to focus on. Their chosen value could be an emotional attachment (such as mental health), a reason why they like going to the beach (such as spending time with friends, going on vacation to the beach, or surfing), cultural or societal significance, tied to food or tradition (such as enjoying ceviche or seafood), or an appreciation of the ocean system (such as wanting to protect marine mammals).
8. Once students have chosen their value they want to model, advance to **Slide 10** and give them their next task, which is to refine their list of components, processes, and interactions. Ask students to choose **10 - 12 items** from their original list that are most connected to their value. They will prioritize including these items in their model.

INVESTIGATE: Building Our Model (30 minutes)

1. Advance to **Slide 11** and tell students that their next task is to build a model that centers their chosen value and includes the 10 - 12 items from their original list that are most closely connected to their value.

Student models will be made in SageModeler and will answer the **modeling question**: How will climate change impact the reason why you care about Southern California's ocean?

2. **Tip for Facilitator:** Before having students make their own model, open the SageModeler template and demonstrate the process by adding (1) a value and (2) one or two items. Condensed steps are included on **Slide 12**, but the modeling process is as follows:

Modeling Process:

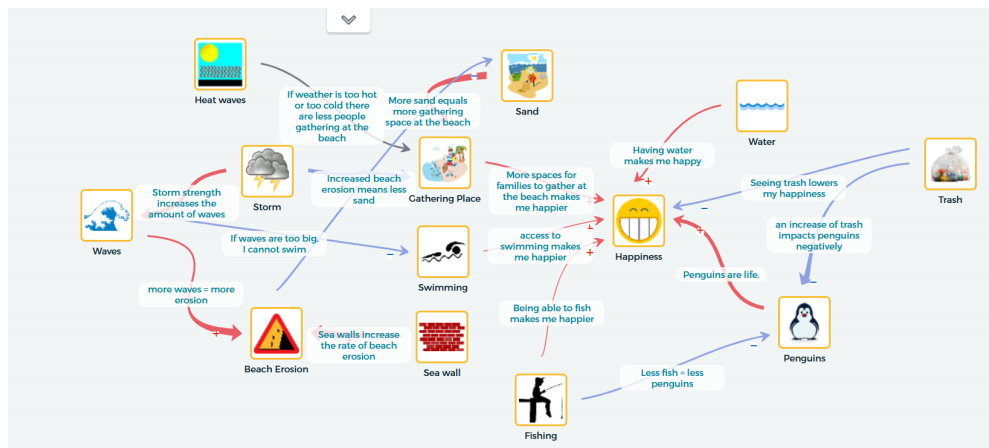
- Begin by opening the [SageModeler template](#).
- Click on the *New Image* icon in the upper right corner and search for an image to represent your chosen value. Add it to your picture break and then drag it into the modeling window. Make sure to label the picture with your chosen value.

- Search for photos to represent the other components, processes, and interactions that you want to include in your model. Add the components, processes, and interactions into your model and label them.
 - Begin connecting the parts of your model together by defining their relationship. To do so, you'll need to click on one icon and then drag the arrow to the second icon that you want to connect to. You'll then be able to double-click on the arrow to define the relationship, which could be:
 - **Positive relationship:** As one factor increases, the second factor also increases.
 - **Negative relationship:** As one factor increases, the second factor decreases.
 - **Varied relationship:** As one factor increases, the second factor can increase or decrease depending on different conditions
 - As you make connections in your model, click on the arrow one last time to add a text box section and describe the relationship using words.
3. These steps (from point 2 above) are also reflected in the slides. Students may reference the modeling process described in the following **Student SageModeler Instructions (Slides 13-20; in Science Journal p.11-18)**:
- **Slide 13 (hidden slide):** Choose one member of your team to **create a copy** of the **SageModeler template** or visit <https://tinyurl.com/AISciModel>. The whole team will work on one model together.
 - **Slide 14:** To start your model, click on the **new Image** icon at the top-left corner of the screen. Choose an image to represent your chosen value and then drag it into the modeling window. Make sure to label the picture with your **chosen value**.
 - **Slide 15 (hidden slide):** Next, search for images to represent the other **components, processes, and interactions** that you want to include in your model, making sure to label all of them.
 - **Slide 16:** To **connect two items** in your model, hover over the first item. An arrow will appear. Click and drag that arrow over to the item you want to connect it to. Creating a causal relationship.
 - **Slide 17:** To **describe the relationship** between two items, double click on the arrow that connects them. Then choose which descriptions best represent the relationship between the two items.
 - **Positive relationship:** As one factor increases, the second factor also increases.
 - **Negative relationship:** As one factor increases, the second factor decreases.
 - **Varied relationship:** As one factor increases, the second factor can increase or decrease depending on different conditions

- Ask students to brainstorm: **How do human activities that drive climate change affect different types of value—and how can one activity cause value to go up, go down, or depend on the situation?**
 - **Slide 18-19:** This slide provides students with examples of different causal relationships that they can model:
 - **Linear:** A direct cause-and-effect relationship where one element or process directly impacts another.
 - **Causal:** A multi-step sequence where an initial event triggers an intermediate change, which then leads to a final outcome.
 - **Feedback Loop:** A circular relationship where the outcome of a process feeds back into the original cause, amplifying (positive feedback) or reducing (negative feedback) the effect over time.
 - **Indirect Loop:** A variable (A) influences an outcome (C) both indirectly through an intermediate variable (B) and through a separate direct pathway.
 - **Slide 20 (hidden slide):** **Continue** to add components, processes, and interactions from your list to the model. Save your model when you're done.
4. If there's time once the model is complete, advance to **Slide 21** and have students click on the *Simulate* icon in the upper right-hand corner of SageModeler to simulate change on different factors within your model. As you make simulations think about the following:
- How does a change in a component, process, or interaction within your model affect your team's chosen value?
5. Advance to **Slide 22** and have students take a snapshot to save an image of the model that will be used in the next module lesson. **Ask students to upload the images to your classroom Google Drive.**

Depending on the type of device available, have students snap a picture. Below are a couple of the built in programs for taking a picture.

- If on a Windows computer
 - Use the print screen button
 - Or use the built in snipping tool platform
- If on a Mac computer
 - Press **Command (⌘) + Shift + 4** at the same time
- If on a Chromebook
 - Press **Ctrl + Shift + Show Windows** at the same time



NAVIGATE: Gallery Walk (25 minutes)

1. As students finish up their models, let them know that their next step will be to conduct a gallery walk to compare their team's model with the models created by other teams.
2. Advance to **Slide 23**. Before students start conducting their gallery walk, inform them that as part of the gallery walk, you would like them to provide feedback on the models of their peers on post-it notes. Each post-it note will represent one portion of feedback.
3. On **Slide 24**, to help guide students through providing feedback, share the following questions that they should consider as they visit each model.
 - Does this model include a variety of components, processes, and interactions? Is there anything that is missing that might help explain the system better?
 - Do the components, processes, and interactions help explain why that value is important and how it might be impacted by climate change?
 - What part of this model helped you understand how climate change is impacting the chosen value?
 - What suggestions do you have to make this model more complete or clear?
4. As students finish up their gallery walk, invite them to take a look back at their model. Ask student teams to answer the following questions as a group.
 - What patterns did you notice as you conducted your gallery walk?
 - Were there any ideas or aspects you saw in other models that you would like to incorporate into yours?

PROBLEMATIZE: Reflecting On What We've Learned (10 minutes)

1. Bring the class back together as you finish up for the day. Tell the students that like before, you would like to take a moment for them to reflect in their Science Journals.
2. Advance to **Slide 25** and ask student teams to reflect on **Module 2.1 Building a Model (Student Reflection), p. 19** of their Science Journals.
 - What did you learn about the components, processes, and interactions in the ocean while building your model?
 - How are these components, processes, and interactions connected to the value you chose?
 - Which parts of your model would you like to learn more about? How might you investigate or gather information to explore these areas further?
3. As students wrap up their discussion, and if time allows, have teams share with the entire class.
4. Thank the students for their time today. Advance to **Slide 26** and let students know that during the next session, they are going to take more time to refine and incorporate new ideas into their model. They have already incorporated ideas from their interviews, but next, they will have a conversation with a generative AI platform to identify potential areas for improvement on their model.